# Rank contraction: the complete measurement boundary

Human Compression project

Input proposal; no factors

The original is Peter A. Brooksbank and Dimitri Leemans, *Rank reduction of string C-group representations*, arXiv:1812.01055v2, 3 May 2019. All six pages are included. The scientific edition has a distinct root review of its 22 source clauses and 20 proof contexts. This input packet uses that acceptance; it does not repeat the mathematical, GAP or Lean review.

| Key | Complete document | Pages |
|---|---|---|
| L037 | Original, including bibliography and identification banner | 6 |
| U | Contraction of string markings through a central factor | 3 |
| A1 | Rank intervals from one string marking | 5 |
| CODE | Complete finite-example program for rank contraction | 1 |

The full common paper is acquired once. Every page of the application, including its final bibliography page, remains part of its cost. No earlier Topic B common paper is a free premise, and no overlap discount is made. A later written comparison must charge $U + A1$; a separate program-inclusive comparison must also charge the complete program. Its printed and raw-file forms are alternative representations and are not added together. A conditional application comparison presupposes acquisition of the entire common paper and the declared complete code.

The one whole GAP file, `check_examples.g`, supplies finite certificates for the literal four-dimensional matrix example and three $A_{11}$ markings. Its doubled-edge orbit checks are also validation. It is not a premise for the uniform contraction theorem or the infinite rank families. The complete one-page appendix prints the whole file, without line selection. The separate scientific driver `check.py` and the complete `RankContraction.lean` file are inventoried and counted separately. That Lean file has four scoped algebraic declarations; it does not formalize string C-groups, the contraction induction, the classical seeds, or the whole original.

The primary coordinate is NFC non-whitespace characters in the complete raw PDF extraction. The policy was fixed before computing the counts. The full 30 coordinates retain five units, NFC and NFKC, and three text variants: raw untrimmed, exact furniture-selected, and calibrated selected. The units are whitespace tokens, alphanumeric runs, non-whitespace characters, characters with collapsed whitespace, and UTF-8 bytes with collapsed whitespace. Printed hyphens remain literal. No factors are computed or adopted here.

## Visible symbols and retained costs

Extraction uses pypdf 6.13.0 and FontTools 4.63.0. Every returned page string is compared with uninstrumented pypdf extraction. The source has 23 actual font dictionaries and 478 used encoded glyph classes. Replacement Unicode fonts are checked through their actual embedded programs, glyph indices and ToUnicode maps. A shared font name does not identify a font: dictionary, effective encoding and original byte events are bound separately.

The source's matrix on page 3 uses eight private-use characters for six visible square-bracket component classes. Their actual embedded outlines were inspected. Calibration replaces each component by its corresponding Unicode top, extension or bottom piece, preserving one visible component per event. It does not merge the pieces into two invented delimiter tokens. No scientific glyph is deleted as whitespace. The raw primary preserves the original extraction unchanged.

Figure 1 on source page 4 has three diagrams. Eight XYCIRC curved-fragment classes occur 33 times each; their literal `/c16` through `/c23` extraction strings remain in every variant. They are not silently replaced by circles or interpreted as complete edges. The diagram's full graphical complexity is unknown. This qualification also applies to the distinction between extractable labels and the arrangement of a matrix, table or graph. All six source pages and all nine replacement pages, together with 27 exceptional or supplementary glyph outlines, were viewed.

Only exact page-number footers and repeated source running headers are removed in the two selected sensitivities. The source's first-page arXiv banner remains included. Titles, abstracts, author blocks, acknowledgements, source questions and references are retained. The programme appendix has no missing lines or pruned ranges; its inclusion proof removes no printable character and does not alter any measured text.

The standard end criterion, the stated highest-rank classical constructions, orthogonal identifications and the prior $A_{11}$ exhaustion remain explicit external mathematical inputs. In particular the complete Monson–Schulte III primary is not acquired by this packet; its exact construction premise is retained through the source. These proof costs, the full graphical cost, historical software cost and a complete formalization have value *unknown*, not zero. The source's rank-gap Conjecture 5.1 remains an open research target; no solution or new question credit is asserted.

The frozen input proposal supplies exact byte bindings, page and glyph-event spans, all 120 PDF coordinate cells and 30 raw-file cells, a portable replay, and corruption controls. A separate independent input review is required before numerical comparison.